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Prenominal Adjective Order and Communication Context

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Will the communication context in which an object is described influence the ordering of prenominal adjectives? While viewing slides of objects in four different contexts, *S*s heard two descriptions of an object. These descriptions contained two adjectives (size and color) in either a normal or an inverted order with either a stress or a list intonation. If the context necessitated a discrimination based on color, then *S*s preferred the inverted order of adjectives If size was the critical attribute, *S*s strongly preferred the normal order relative to control contexts. These relations were obtained only with stress intonation. The results supported a pragmatic-communication rule which determined prenominal adjective ordering rather than a simple syntactic or semantic rule.

English speakers order prenominal adjectives according to some rule or convention: the phrase *the large red car* is more acceptable than *the red large car*. This fact suggests that adjective order is not determined in arbitrary or idiosyncratic fashion; however, the order in which they occur cannot be accounted for by a simple syntactic rule (Martin, 1969; Danks & Glucksberg, 1971). What then are the factors which constrain the order of prenominal adjectives?

Martin (1969) concluded that the ordering of a string of prenominal adjectives could best be predicted by the definiteness of denotation and the absoluteness of the adjectives. The more absolute or definite the adjective, the stronger the tendency to place it closer to the noun. Definiteness and absoluteness seemed to define the same dimension since they were highly correlated, $r \geqslant .90$. Absoluteness was defined by Martin in terms of the number of ". . . comparisons you would have to make among a class of objects before you could choose . . . the adjectives" (1969, p. 700). In the phrase *the large red car*, *red* is more absolute than *large* because one would need to make no comparisons between cars to decide if a given one was red but would need to compare at least two cars of different sizes to determine if one was large. The more constant an adjective's meaning across different nouns, the more definite it is according to Martin. In the above example, the meaning of *red* would remain constant whether it were modifying *car* or *barn*, but the meaning of *large* would not.

Danks and Glucksberg (1971) have suggested that Martin's absoluteness and definiteness dimension(s) can be incorporated into a dimension of intrinsicalness Adjectives which denote properties most intrinsic to the noun are placed closest to the noun. Hence, adjectives denoting place of origin should appear closest to the noun, preceded by adjectives denoting superficial properties such as color. The color adjectives would in turn be preceded by size adjectives since size is determined in relation to other objects, rather than being inherent in the object. Scaling of adjective orders supported this interpretation.

Danks and Glucksberg viewed the intrinsicalness rule for adjective ordering as the most frequent case of a more general pragmatic-communication rule. The more intrinsic the relation between adjective and noun, the less likely the adjective will be used to discriminate the noun referent from others in its subcategory. Hence, adjectives which do discriminate tend to occur first in a string of adjectives. The communication context in which the referent was being identified would

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determine which adjective was most discriminating. If the speaker knew that the listener would have to discriminate between two red automobiles, one large and one small, then the normal adjective order would be optimal, for example, *the large red car*. But, if the discrimination was between a red car and a blue car, both of them large, then the speaker might well say, *the red large car*. The more discriminating adjective is moved to the primary position in order to enhance the probability of the listener making the correct identification. Inversion of adjective order is also accompanied by a change in intonation. In the normal order, stress on the first adjective and a juncture following is optional, but the stress and juncture are obligatory for the inverted order. Martin (1970) found that *S*s placed juncture following adjectives that were not in normal order.

The present experiment tested this pragmatic-communication rule by having *S*s choose between normal and inverted adjective orders in a variety of contexts. In a pilot study *S*s were permitted a free verbal response. However, *S*s tended to omit nondiscriminating adjectives, thereby yielding no order information, or they used circumlocutions which were uninterpretable vis-à-vis adjective orders, for example, *individual transportation* for car and *lead containing writing instrument* for pencil. Hence, we devised a forced-choice procedure. Intonation was also varied between a stress-juncture presentation and a list presentation of the phrases.

METHOD

Subjects

Fifty female undergraduates enrolled in the introductory psychology course at Kent State University served as *S*s in the experiment. They were given points which were convertible to course credit for their participation. All were native English speakers and were naive to this experimental situation.

Materials

Colored line drawings of 4 versions of 10 different objects were used to construct 80 slides. The objects were a car, a pencil, a key, a clock, a cup, a button, a fish, a hammer, scissors, and a table. Each object was drawn in two different sizes and two different colors in order to obtain four unique versions of the object. The two sizes of each object differed in more than a simple magnification or reduction in size. They also differed on other characteristics, for example, the large car was a sedan and the small car was a sports car.

Eighty different slides were composed from the forty drawings. Each slide contained four different drawings, one in each quadrant of the slide. The object in the upper-left hand corner was designated the criterion object. The 80 slides were divided into four different slide types representing four different experimental conditions. See Table 1 for examples.

TABLE 1

EXAMPLES OF THE COMPOSITION OF THE FOUR SLIDE TYPES

Both Relevant		Neither Relevant	
LARGE	small	LARGE	small
RED	green	RED	gold
CAR	car	CAR	fish
small	large	large	small
red	green	blue	yellow
car	car	table	pencil
Color Relevant		**Size Relevant**	
LARGE	small	LARGE	small
RED	brown	RED	red
CAR	scissors	CAR	car
small	large	large	small
gold	green	blue	gold
key	car	table	clock

Two groups of slides represented control contexts for the experiment. It was hypothesized that when both adjectives were necessary to discriminate the criterion object from the others (Both Relevant condition) or when neither adjective was necessary for the discrimination (Neither Relevant condition), then the normal adjective order would be preferred. However, this preference would probably not reach 100%. Ten Both Relevant slides each depicted the four versions of a single object. Each of 30 Neither Relevant slides depicted four objects from four different object sets. Distinguishing the criterion object from the others in the context of Both Relevant slides would require the use of both color and size adjectives, but in the context of Neither Relevant slides, neither adjective would be necessary. Hence, the normal order of adjectives would be preferred in both conditions.

Color Relevant and Size Relevant slides provided the experimental contexts. The 20 Color Relevant slides depicted two objects from the same object set differing from each other only on the color dimension, plus two objects from two of the remaining nine sets. Twenty Size Relevant slides depicted two objects from a single set differing from each other only on the size dimension plus two irrelevant objects. Since Color Relevant provided a context in which color was the discriminating attribute, it was hypothesized that the inverted adjective order would be preferred more often than in either the Both Relevant or Neither Relevant control conditions. With Size Relevant, where the size adjective was the discriminating attribute, the preference for normal adjective order should be even stronger than in the control contexts.

In the 80 slides each colored line drawing appeared twice as a criterion object, but never twice in a particular slide type. In color Relevant and Size Relevant slides, the object contrasting with the criterion was placed equally often in the other three quadrants of the slide.

Test construction and procedure

Ten Neither Relevant slides were randomly selected as warm-up slides The remaining seventy slides followed in a modified random order· No two slides with criterion objects from the same object set followed in succession; no more than two slides with criterion objects of the same color occurred in sequence; and no more than three slides with criterion objects of the same size were presented in sequence.

The slides were presented to groups of *S*s at 15 sec intervals. After viewing the slide for five secs, *S*s heard two tape-recorded descriptions of the criterion object. The descriptions always consisted of a determiner, two adjectives, and a noun. In one description adjective order was Normal, for example, *the large red car*, and in the other it was Inverted, for example, *the red large car*. Whether the normal or inverted order was heard first or second was counterbalanced across all conditions. The *S*s were instructed to select the phrase which would better ensure a listener identifying the criterion object from the set. They circled a "1" or "2" on an answer sheet to indicate their choice.

Intonation was a between-*S* variable The Stress intonation group ($N = 25$) heard phrases with heavy stress on the first adjective, a juncture between the two adjectives, but no juncture between the second adjective and noun, for example, *the RED // large car*. The List intonation group ($N = 25$) heard phrases with stress on both the adjectives and the noun and junctures between each word. The stress was not as strong nor the junctures as large in the List condition as in the Stress one. The intonation patterns were used regardless of the order of adjectives (Normal or Inverted).

Results

The proportion of preference for the inverted adjective order for each slide type for each *S* was calculated. This measure was further divided to compare preference for normal order when the normal order was the first description heard and when it was the second. Proportions rather than frequencies were used since there were only 10 Both Relevant slides and 20 each of the others. The 10 warm-up trials were excluded from the analysis.

There were significant main effects of intonation, $F(1,48) = 14.88$, $p < .001$, and slide type, $F(3,144) = 28.12$, $p < .001$. But these effects can be interpreted only with respect to their interaction, $F(3,144) = 22.10$, $p < .001$. The means for this interaction are presented in Table 2. Although the pattern of means in the Stress and List conditions was the same, the differences within the Stress condition were much larger. Newman–Keuls comparisons showed that there were no differences within the List condition that approached significance. However, within the Stress group both Color Relevant and Size Relevant preferences were significantly different from both controls (Both Relevant and Neither Relevant), $p < .01$ for all comparisons The two controls did not differ significantly from each other

TABLE 2

Mean Proportion of Inversions Preferred as a Function of Slide Type and Intonation Pattern

	Relevant adjective			
Intonation	Both	Neither	Color	Size
Stress	.168	.156	.574	.036
List	.068	.072	.090	.054

In order to confirm the analysis of variance results, a sign test was used to compare the number of inversions preferred in the experimental contexts (Color Relevant and Size Relevant) to a control context. Only Neither Relevant was used for the control comparison since Both Relevant was tested only half as frequently. For the Stress group, 18 of 18 *S*s (7 *S*s had no difference in sign) preferred more

inversions for Color Relevant than Neither Relevant and 11 of 15 *S*s (10 *S*s had no difference) preferred fewer inversions for Size Relevant than Neither Relevant. These differences were significant, $p < .001$ and $p < .06$, respectively. For the List group, only 5 of 7 *S*s (18 *S*s had no difference) preferred more inversions for Color Relevant and 3 of 6 *S*s (19 *S*s had no difference) preferred fewer inversions for Size Relevant when compared to the control slides. These differences for the List group did not approach significance, $p > .20$ for both.

Although the main effect of intonation was significant, the differences were due almost entirely to the preference for the inverted order to Color Relevant slides by Stress *S*s. The preferences to the other slide types did not differ between the two groups.

The effect of the inverted adjective order occurring first or second on a trial did not approach significance, $F(1, 48) < 1$. However, the interaction of order of occurrence with intonation was significant, $F(1, 48) = 5.22$, $p < .03$. The Stress group had a slight tendency to prefer the inverted order more when it was presented second: means are .22 for first, .24 for second. The reverse tendency was true for the List group: means are .08 for first, .06 for second. None of the other interactions was significant.

Discussion

At first blush, the ordering of prenominal adjectives would seem to be a function of some syntactic rule. Both Annear (1964) and Vendler (1963) attempted to state such a rule in strict syntactic terms, but their effort was unsuccessful. Martin (1969) and Danks and Glucksberg (1971) proposed semantic rules to explain the adjective ordering. Their semantic rules were closely related and were confirmed by their scaling studies However, there were some minor anomalies in the Danks and Glucksberg data which led them to suggest that the context influences the scaling. The results they obtained and also Martin's results may represent the most frequent case of a more general rule. That more general rule is the pragmatic-communication rule that we have described, namely, that the more discriminating adjectives tend to be placed first in a series of adjectives.

The results for the Stress group completely supported our analysis of the pragmatic-communication rule. When both adjectives were required to identify the correct referent and when neither adjective was required, the normal order of adjectives was preferred nearly 85% of the time. If color was the sole discriminating attribute, then the inverted order was preferred (57%). On the other hand, if the discrimination was based on size alone, the normal order was preferred even more than in the control cases (96%). This latter result was predicted, but unexpected, because of the potential ceiling in the control conditions. That the normal order was not preferred 100% for control contexts implies that the normal order is not as absolute as one might suspect, lending further support to our claim that adjective ordering is not determined solely by a single syntactic rule.

A central tenet of a theory of semantics developed by Olson (1970) holds that for a complete understanding of utterances they must be considered in relation to alternative referents. He defined meaning as "information that permits a choice or differentiation among alternatives" (1970, p. 265). The larger the context or number of alternatives, the greater the amount of information contained in the utterance. With only a few alternatives the speaker can specify precisely the intended referent with a minimum of verbiage. But, if the alternative referents available to the listener are not immediately obvious, then the speaker must in effect rule out the remainder of the world, excepting only the referent. Such was the problem with our pilot study in which a free response was permitted The alternative nonreferents constituted such a small set that *S*s produced only minimal descriptions. The present experiment lends support for Olson's theory in that it demonstrates that *S*s are

sensitive to the alternative context and that they structure their utterances accordingly.

Although our *S*s did not actually produce utterances, the task can be considered an analogue of the speaker's process. The *S*s were instructed to select the better phrase for communicating to a listener by placing themselves in a speaker's role. Other studies have found that speakers will modify syntactic patterns depending on the context. Prentice (1967) found that when *S*s were cued by a particular object, they produced either an active or passive sentence such that the name of the cuing object was the subject The *S*s tended to produce an active sentence if the cuing object was the agent of an action and a passive sentence if it was the recipient. Similar results were obtained by Tannenbaum and Williams (1968). Hence, speakers changed the voice of their descriptive sentence so that the focused object occupied the primary sentence position.

In the present experiment, *S*s placed the more important adjective first. Hence, we seem to have further confirmation of a general psychological principle that important information occupies a primary position in communication. However, there is at least one major difference between our study and those investigating sentence voice Inverted adjective order appears to violate a relatively well-established grammatical rule. Changing a sentence from active to passive is merely a stylistic variation. We might conclude that our Stress *S*s produced over 50% ungrammatical utterances for Color Relevant. However, an alternative interpretation is that the intonation pattern interacts with the syntax to determine acceptability. Danks and Glucksberg (1970) have shown that acceptability and grammaticalness are not independent of each other. There are "normal, grammatical" sentences which must be accompanied with the appropriate intonation before they will be considered acceptable, for example, deeply embedded sentences and questions. Hence, it should not be too surprising that the syntactic ordering of prenominal adjectives, for which a motivated, purely syntactic rule cannot be discovered, must be cued by appropriate intonation patterns.

Another factor contributing to the importance of intonation is that prenominal adjectives constitute a left-branching surface structure. It has been well established that left-branching structures are difficult for the listener to decode (for example, Stolz, 1967). The intonation patterns provide the listener with additional cues to facilitate processing. To a large extent these comments on the function of the stress intonation pattern are somewhat speculative. To confirm them a study in which the effect of the various adjective orders and intonation patterns on the listener's processing of the noun phrases is determined directly is required.

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